Supplementary material

Two de novo design routes, one natural topology: Convergence of SCUBA designs and an RFdiffusion-derived binder

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Figure S1 and Tables S1A, S1B, S2 and S3 document the structural comparisons and database-search summaries reported in the manuscript. Coordinate inputs, complete search outputs and analysis scripts are available in the v1.1.2 archive at https://doi.org/10.5281/zenodo.23270724.

Figure S1

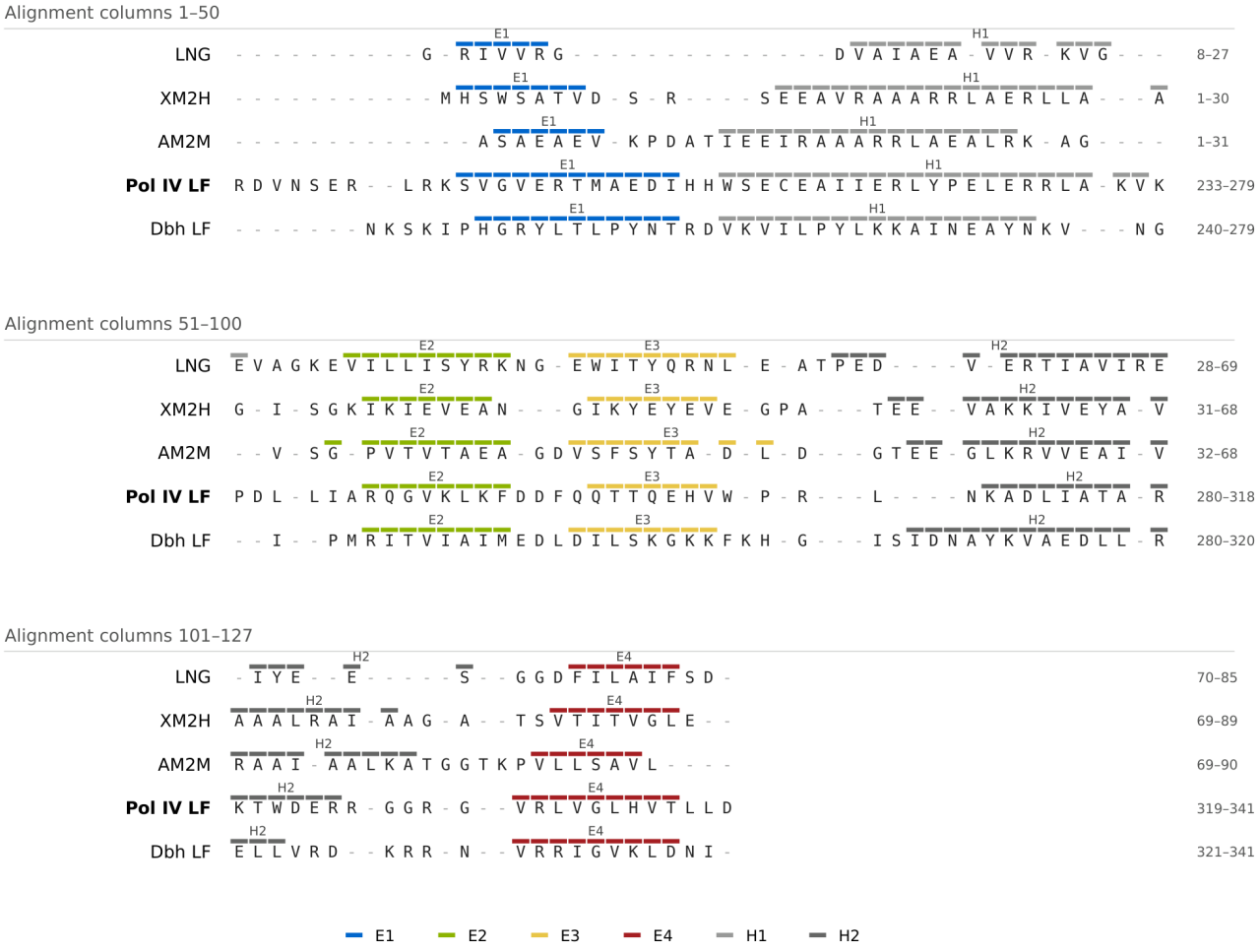


Figure S1. Structure-guided sequence comparison of designed proteins and natural little-finger domains. The four frozen pairwise TM-align alignments were combined using Pol IV LF (4R8U:A, residues 233–341) as the common reference. Rows show LNG, XM2H, AM2M, Pol IV LF and Dbh LF; dashes indicate alignment gaps. Bars mark the principal secondary-structure elements defined in Table S1A: strands E1–E4 are blue, green, yellow and red, respectively, and helices H1 and H2 are grey. Elements are numbered in polypeptide-chain order. Numbers at the right indicate the first and last PDB residue represented in each displayed block. Insertions relative to Pol IV LF are left-aligned within shared gap slots for display; their placement does not establish residue equivalence between the other proteins. Pairwise structural statistics and sequence identities are reported in Table S2.

Table S1A Secondary structure element boundaries

Principal strands E1–E4 and helices H1–H2 are numbered in polypeptide-chain order. Boundaries use original PDB residue numbering. DSSP assignments were mapped by chain, residue number and insertion code; principal strands and helices required at least three and four residues, respectively.

Protein	Element	Start residue	End residue	Residues
LNG	E1	9	13	5
LNG	H1	16	28	13
LNG	E2	34	42	9
LNG	E3	45	53	9
LNG	H2	57	74	18
LNG	E4	78	83	6
XM2H	E1	2	8	7
XM2H	H1	13	30	18
XM2H	E2	36	42	7
XM2H	E3	45	51	7
XM2H	H2	57	76	20
XM2H	E4	82	88	7
AM2M	E1	2	7	6
AM2M	H1	13	28	16
AM2M	E2	34	42	9
AM2M	E3	45	53	9
AM2M	H2	57	77	21
AM2M	E4	84	89	6
Pol IV LF	E1	243	254	12
Pol IV LF	H1	257	278	22
Pol IV LF	E2	286	293	8
Pol IV LF	E3	298	304	7
Pol IV LF	H2	310	324	15
Pol IV LF	E4	330	338	9
Dbh LF	E1	246	256	11
Dbh LF	H1	259	275	17
Dbh LF	E2	283	290	8
Dbh LF	E3	294	301	8
Dbh LF	H2	308	323	16
Dbh LF	E4	331	339	9

Table S1B Strand pairing measurements

Contact counts represent interstrand Cα atom pairs separated by less than 5.5 Å, not hydrogen-bond counts. Minimum distances are the closest interstrand Cα separations. Axes run from the first to the last Cα atom of each strand; negative axis cosines indicate opposing directions. All five structures have sheet order E1–E4–E2–E3, with the three adjacencies antiparallel.

Protein	Strand pair	Sense	Cα pairs <5.5 Å	Minimum Cα distance (Å)	Axis cosine
LNG	E1-E4	antiparallel	5	4.5	-1.00
LNG	E2-E3	antiparallel	8	4.2	-0.92
LNG	E2-E4	antiparallel	6	4.0	-0.95
XM2H	E1-E4	antiparallel	9	4.2	-0.96
XM2H	E2-E3	antiparallel	8	4.3	-0.97
XM2H	E2-E4	antiparallel	7	4.6	-0.95
AM2M	E1-E4	antiparallel	8	4.2	-0.97
AM2M	E2-E3	antiparallel	11	4.2	-0.93
AM2M	E2-E4	antiparallel	8	4.3	-0.93
Pol IV LF	E1-E4	antiparallel	10	3.8	-0.97
Pol IV LF	E2-E3	antiparallel	7	3.8	-0.96
Pol IV LF	E2-E4	antiparallel	11	4.1	-0.95
Dbh LF	E1-E4	antiparallel	11	3.6	-0.96
Dbh LF	E2-E3	antiparallel	8	3.8	-0.96
Dbh LF	E2-E4	antiparallel	9	4.4	-0.93

DSSP-omitted residues were assigned coil for segmentation. In full-chain Dbh, residue A35 was omitted by DSSP; it lies outside the analyzed little-finger domain. The residue-key mapping prevents this omission from shifting the domain assignments.

Table S2 Pairwise structural comparisons

Sixteen TM-align comparisons comprise ten unique pairs among the five analyzed structures and six comparisons of designed proteins against full natural polymerase chains. qTM and tTM are normalized by query and target length, respectively, with the corresponding length-dependent d_0 parameters; these are two normalizations of the same alignment. RMSD and sequence identity refer to aligned residues, not whole-sequence identity.

Query	Query origin	Query extent	Target	Target origin	Target extent	Aligned residues	RMSD (Å)	Identity (%)	qTM	tTM
Pol IV LF	natural	domain	Dbh LF	natural	domain	95	1.91	11.6	0.737	0.780
LNG	RFdiffusion	domain	Dbh	natural	full chain	71	3.82	1.4	0.492	0.164
LNG	RFdiffusion	domain	Dbh LF	natural	domain	70	3.93	10.0	0.476	0.405
LNG	RFdiffusion	domain	Pol IV	natural	full chain	72	3.70	8.3	0.509	0.171
LNG	RFdiffusion	domain	Pol IV LF	natural	domain	72	3.70	8.3	0.509	0.415
AM2M	SCUBA	domain	Dbh	natural	full chain	86	2.86	9.3	0.668	0.220
AM2M	SCUBA	domain	Dbh LF	natural	domain	86	2.86	9.3	0.668	0.609
AM2M	SCUBA	domain	Pol IV	natural	full chain	83	2.50	10.8	0.660	0.217
AM2M	SCUBA	domain	Pol IV LF	natural	domain	83	2.50	10.8	0.660	0.573
XM2H	SCUBA	domain	Dbh	natural	full chain	88	2.65	11.4	0.687	0.227
XM2H	SCUBA	domain	Dbh LF	natural	domain	88	2.65	11.4	0.687	0.623
XM2H	SCUBA	domain	Pol IV	natural	full chain	86	2.91	9.3	0.629	0.217
XM2H	SCUBA	domain	Pol IV LF	natural	domain	86	2.91	9.3	0.629	0.548
LNG	RFdiffusion	domain	XM2H	SCUBA	domain	76	2.85	9.2	0.633	0.579
AM2M	SCUBA	domain	LNG	RFdiffusion	domain	74	3.14	12.2	0.524	0.575
XM2H	SCUBA	domain	AM2M	SCUBA	domain	87	1.81	40.2	0.800	0.792

The query and target labels reproduce the archived table. For the designed proteins, “domain” denotes the analyzed single-domain coordinate input. The natural reference alignment has 95 aligned residues, RMSD 1.91 Å and identity 11.6%; its TM-scores 0.737 and 0.780 reflect different normalization lengths, not an uncertainty interval.

Table S3 Database search summaries

Summaries were calculated from the archived, headerless Foldseek TSV files. qTM is the query-normalized TM-score. Source filenames identify files in results/foldseek/ of the v1.1.2 archive. These counts describe returned retrieval neighbourhoods rather than formal classification of the designs.

A CATH50 superfamily counts

Query	Superfamily	Hits qTM ≥0.5	Total qTM ≥0.5	Source TSV
LNG	3.30.950.10	6	9	9bk5A_cath.tsv
LNG	3.30.1490.100	2	9	9bk5A_cath.tsv
LNG	3.30.70.600	1	9	9bk5A_cath.tsv
XM2H	3.30.1490.100	21	24	xm2h_cath.tsv
XM2H	3.90.70.120	1	24	xm2h_cath.tsv
XM2H	3.30.470.20	1	24	xm2h_cath.tsv
XM2H	3.60.20.10	1	24	xm2h_cath.tsv
AM2M	3.30.1490.100	14	14	am2m_cath.tsv
Pol IV LF	3.30.1490.100	37	37	4r8u_LF_cath.tsv
Dbh LF	3.30.1490.100	35	35	1k1s_LF_cath.tsv

CATH50 v4.3.0. Superfamilies were mapped using cath_target_superfamilies.tsv. LNG's nine qualifying hits span topology groups 3.30.950, 3.30.1490 and 3.30.70 within architecture 3.30.

B AlphaFold DB Swiss Prot retrieval

Query	Returned rows	Rows ≥0.5	Rows ≥0.6	Best qTM	Unique ≥0.5	Shared control	Source TSV
Pol IV LF	779	268	268	0.9965	258	258	4r8u_LF_afdbsp.tsv
XM2H	571	175	4	0.6991	172	148	xm2h_afdbsp.tsv
AM2M	272	48	2	0.6068	47	46	am2m_afdbsp.tsv
LNG	690	7	0	0.5372	7	3	9bk5A_afdbsp.tsv

AlphaFold DB Swiss-Prot v6 (2025_03). Thresholds refer to qTM. "Unique ≥0.5" counts unique UniProt accessions among qualifying rows. "Shared control" is the intersection with the Pol IV LF accession set at qTM ≥0.6. Returned-row counts and unique-accession counts differ because multiple target records can represent one accession; 268 qualifying control rows represent 258 unique accessions.

Table S3 Database search summaries continued

C Selected LNG hits in PDB100 and CATH50

Database	Target identifier	Returned rows	qTM rank	qTM	tTM	Probability	Source TSV	Original row
PDB100	4r8u-assembly2_B	4911	197	0.4145	0.1491	0.887	9bk5A_pdb.tsv	9
PDB100	1k1s-assembly1_A	4911	590	0.3437	0.1284	0.150	9bk5A_pdb.tsv	255
CATH50	af_Q9VNX1_325_451_3.30.1490.100	580	5	0.5385	0.3771	0.996	9bk5A_cath.tsv	1

Ranks are one-based ordinal positions after sorting all returned rows by descending qTM, retaining original row order for ties. “Original row” is the one-based line number in the source TSV. The PDB100 targets are whole polymerase chains; the CATH50 target is a different protein’s little-finger-superfamily domain. Database composition, matched proteins and search settings differ, so this comparison does not isolate domain resolution. PDB100 used PDB_DATE 250101 and exhaustive search; CATH50 used v4.3.0. All searches used -e 10.

D LNG qualifying AlphaFold DB accessions

UniProt accession	Target identifier	qTM	Probability	Shared with control	Source TSV	Original row
Q7USY7	AF-Q7USY7-F1-model_v6	0.5084	0.975	Yes	9bk5A_afdbsp.tsv	3
Q93RN4	AF-Q93RN4-F1-model_v6	0.5308	0.855	No	9bk5A_afdbsp.tsv	19
Q3ZWB2	AF-Q3ZWB2-F1-model_v6	0.5372	0.855	Yes	9bk5A_afdbsp.tsv	23
Q9JWJ7	AF-Q9JWJ7-F1-model_v6	0.5093	0.817	No	9bk5A_afdbsp.tsv	28
A5FSS1	AF-A5FSS1-F1-model_v6	0.5142	0.772	Yes	9bk5A_afdbsp.tsv	52
P27475	AF-P27475-F1-model_v6	0.5103	0.601	No	9bk5A_afdbsp.tsv	108
P21636	AF-P21636-F1-model_v6	0.5179	0.301	No	9bk5A_afdbsp.tsv	218

The seven qualifying LNG hits include three accessions shared with the Pol IV LF control at qTM ≥0.6: Q7USY7, Q3ZWB2 and A5FSS1. None of the LNG hits reaches qTM ≥0.6. Source identifiers are retained for traceability; shared accessions do not establish evolutionary homology of the design.